<u>Log Book</u>

Class & Group	S2-08, Group F
Member Names	**<u>Rachel Foo, Kam Yau Shing</u>**
Title	**<u>Investigation on the bacteria on different surfaces in the school compound</u>**

Project planning:

- As you plan your science project, use your lab notebook to capture the questions you hope to investigate, your hypothesis, and your variables.

Research Question	"How does the frequency of a surface being touched affect the amount of bacteria on it?"
Research Hypothesis	H1:"When a surface is touched more frequently, the number of bacteria (estimated) will also increase exponentially."
Independent Variable	The independent variable is the areas the bacteria are taken from.
Dependent Variable	The dependent variable is the number of bacteria and the type of bacteria
Control Variables	a) The swabbing day should be the same day. b) Growing period should be the same time. (e.g 24 hours) c) How to get a sample (e.g dilution)

Note. Time +/- 2 minutes

Log 1: Materials: ● Did you make any changes or modifications to your materials used? ● If yes, state the changes. ● What have the changes affected the experiment?	
Date	T1W5 - 31 Jan - Tuesday
Details	Yes. The experiment no longer requires a ● Light Microscope ● 24x sticker labels ● 8x Microscope Slides with frosted edge ● 8x inoculation loops ● Microscope digital camera ● 25ml of Immersion oil ● 100 ml of 95% Methanol ● 100ml hucker's crystal violet dye ● 100ml iodine solution ● 100ml safrannin O ● 200ml acetone-alcohol solution ● 100ml sterile water ● Light Microscope

Log 2: Experimental design: ● Did you make any changes or modifications to your experiment design? ● If yes, state the changes. ● What have the changes affected the experiment?	
Date	T1W5 - 31 Jan - Tuesday
Details	No. The experiment will go on as designed

Log 3: Procedures: ● Did you make any changes or modifications to your procedures? ● If yes, state the changes. ● What have the changes affected the experiment?	
Date	T1W5 - 31 Jan - Tuesday
Details	Yes. A gram stain will no longer be carried out as the bacteria on the surfaces are unknown.

Log 4:	
Date	T1W5 - 31 Jan - Tuesday
Details	We have swabbed the surfaces: - S208 Table (Cleaned, Control) - S208 Wall - S208 Floor - S208 Door Handle - S208 Door - S208 Table - Canteen Table - Canteen Seat And plated them onto the agar plates. The plates went into the incubator oven at 37°C at 1504 UTC + 8.

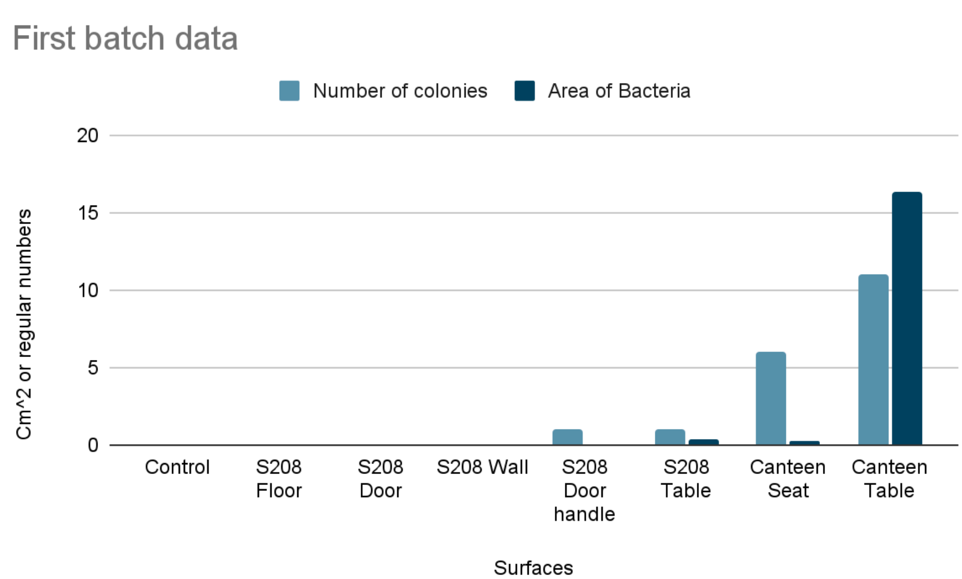
Log 5	
Date	T1W5 - 1 Feb - Wednesday
Details	We have taken out the agar plates at 0810 UTC + 8 to check on the growth of bacteria. As most of the agar plates still did not show signs of growth, we were unable to proceed with the experiment and the plates went back into the incubator oven at 0817 UTC + 8. At 1421 UTC + 8, we have taken out the bacteria colonies again, to check on the progress of the growth of bacteria. As most of the agar plates still did not show signs of growth, we were unable to proceed with the experiment and the plates went back into the incubator oven at 1425 UTC + 8.

Log 6	
Date	T1W5 - 3 Feb - Friday
Details	The agar plates were taken out before our class and we do not know when they were taken out and placed into a fridge. However, we still had to continue so we took pictures of the first batch of bacteria. Additionally, we swabbed - S208 Table (Cleaned, Control) - S208 Wall - S208 Floor - S208 Door Handle

	- S208 Door - S208 Table - Canteen Table - Canteen Seat And plated them, before placing them into the incubator oven at 0852 UTC + 8.

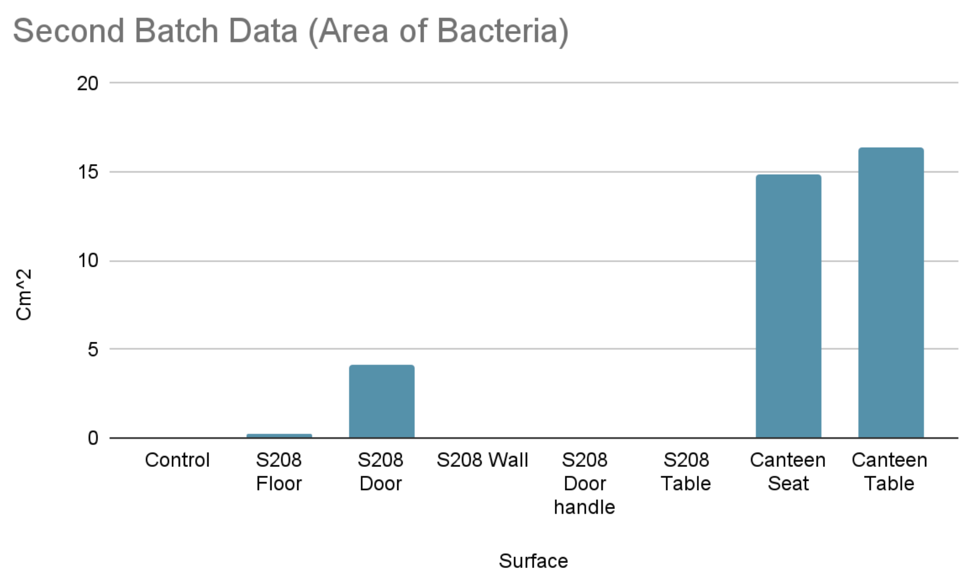
Log 7 Data collection 1. ● Insert your table(s) and graph(s) below. ● Highlight the data points that are anomalous and try to explain why.	
Date	T1W5 - 3 Feb - Friday
Details	First batch data:

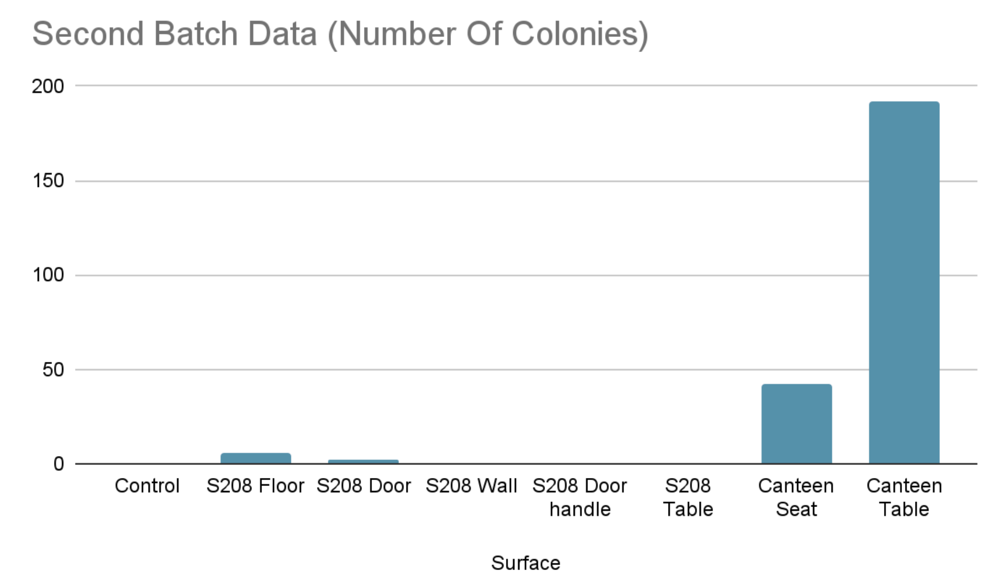
	Number of colonies	Area size of colonies (cm^2, 2d.p)
Control	0	0.00
S208 Floor	0	0.00
S208 Door	0	0.00
S208 Wall	0	0.00
S208 Door handle	1	0.06
S208 Table	1	0.39
Canteen Seat	6	0.24
Canteen Table	11	16.38



Log 8 Data collection 2. ● Repeat the experiment and show the new table(s) and graph(s).	
Date	T1W6 - 6 Feb - Monday
Details	The agar plates were taken out before our class, and we do not know when they were taken out and placed into a fridge. However, we still had to continue, so we took pictures of the second batch of bacteria.

	Number of colonies	Area size of colonies (cm^2, 2d.p)
Control	0	0.00
S208 Floor	6	0.21
S208 Door	2	4.15
S208 Wall	0	0.00
S208 Door handle	0	0.00
S208 Table	0	0.00
Canteen Seat	42	14.84
Canteen Table	192	5.60



Log 9 Data analysis. In point form, ● What are your key findings? ● Limitations and recommendations of your experiment. ● Explain your experimental results using your theory. ● Compare your graphs or histograms with your hypothesis graph and conclude whether your hypothesis is accepted, rejected, or partially accepted. ● Is there a modified theory behind this behavior?	
Date	T1W6 - 7 Feb - Tuesday
Details	● Canteen tables had the most and biggest size colonies (203 colonies, 21.98 cm2) ● Canteen seat had the second highest amount of colonies (48 colonies, 15.08 cm2) ● Other surfaces had fewer colonies (S208 floor: 6 colonies, 0.21 cm2; S208 door: 2 colonies, 4.15 cm2; S208 door handle: 1 colony, 0.06 cm2; S208 table: 1 colony, 0.39 cm2; S208 wall and control: 0 colonies, 0.00 cm2) ● Hypothesis partially accepted: highly touched surfaces (canteen tables and seats) had more colonies ● Door and door handle had fewer colonies than expected ● Results not different from hypothesis

<table>
<tr><td colspan="2">Log 10 Discussion. In point form,

What are the contributions of your research?
What are the practical applications?
What are the areas for further study?
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<tr><td>Date</td><td>T1W6 - 9 Feb - Thursday</td></tr>
<tr><td>Details</td><td>

Provides information on the presence of bacteria on different surfaces in a school compound

Important for promoting hygiene and reducing the spread of infections.
Provides insight into how the frequency of a surface being touched affects the number of bacteria on it.

Basis for developing interventions to reduce bacterial contamination in schools.
Findings of this research can be used to inform the janitors on which surfaces need to be cleaned more frequently or disinfected.
Focus on the impact of different cleaning and disinfection methods on reducing bacterial contamination.
More in-depth research can be done on the different types of bacteria found on different surfaces in a school compound.

Gram stain

Effects of different weather conditions and seasons on bacterial contamination in schools.

Overall, we think the experiment was a fun and enriching experience for the both of us. Not only were we researching on a topic we had great interest in, we learnt quite a number of things during the experiment. For example, learning where there are areas with high amounts of bacteria and learning how to swab areas. When the term ends, we will definitely miss the times we had fun during the experiment. We learnt how to write a report, the scientific method, carry out the experiment, and many more!
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